

Inter-annual changes of partial pressure of CO₂ in the Yangtze River estuary and its adjacent waters

Abstract: In recent decades, with the vigorous development of domestic industry and agriculture, a large amount of inorganic / organic carbon nutrients are imported into the ocean every year in the Yangtze River, so the different distribution of $p\text{CO}_2$ in the Yangtze Estuary will occur in different years, seasons, and even during the day and night. The distribution of $p\text{CO}_2$ in the Yangtze River estuary and its adjacent areas is generally affected by the following processes: organic carbon or carbonate production / degradation / emission, vertical mixing, biological action, runoff input, temperature and salinity changes, etc. This article first divides the Yangtze River estuary into six separate research areas based on the environmental background, and analyzed the inter-annual and seasonal distribution of $p\text{CO}_2$ in each divided research area by underway, algorithm reanalysis, and model data in recent years. Then the main control elements of $p\text{CO}_2$ in the study area are analyzed based on the model and the reanalysis data' prediction elements, combined with the actual environmental background. Through data matching, the relevant elements are combined with the measured $p\text{CO}_2$ to establish corresponding algorithms for different regions and seasons. The inter-annual change of $p\text{CO}_2$ estimated by the algorithm found that as a whole the regional average value of $p\text{CO}_2$ during 1998-2019 showed a slow upward trend and reached the maximum value in 2018, and every 4-5 years, there will be a continuous slow decline and a rapid rise. According to the change trend, it is expected that the annual average of $p\text{CO}_2$ during around 2020-2022 may still decline more slowly, and rapid growth will occur again in the subsequent years 2023-2024. At the end of this article, we have retrieved the climatological distribution of $p\text{CO}_2$ throughout the study area. The study further demonstrated the distribution and variation characteristics of $p\text{CO}_2$ in the Yangtze River estuary and its adjacent waters, and made a preliminary prediction of the inter-annual change of $p\text{CO}_2$.

Key words: Yangtze River estuary, $p\text{CO}_2$, data matching, algorithm's establishment